

Write $\rho^j = R^j - u_h^j$. The stated elliptic reconstruction estimates give $\|\rho^0\|_{\infty, \Omega} \leq \eta_{\text{ell}}^0$, $\|\rho^M\|_{\infty, \Omega} \leq \eta_{\text{ell}}^M$, and $\|\delta_t \rho^j\|_{\infty, \Omega} \leq \eta_{\text{ell}, \delta_t}^j$ for $1 \leq j \leq M$. On each I_j , let $\tilde{R}(s) = R^j + (s - t_j)\delta_t R^j$. The backward Euler equation and the definition of ψ_h^j imply $\psi_h^j = -\delta_t u_h^j$. Consequently, for $w = u - \tilde{R}$, the defining equation for the reconstructions yields, on I_j ,

$$\begin{aligned} \mathcal{K}w(s) &= f(s) - \delta_t R^j - \mathcal{M}\tilde{R}(s) \\ &= f(s) - f^j - \delta_t \rho^j - (s - t_j)\delta_t(f^j + \psi_h^j). \end{aligned}$$

In particular, the last term includes $\delta_t f^j$.

Set $\Phi(s) = \phi_{0, \Gamma}(T - s) = \kappa_0 e^{-\gamma(T-s)}$. Applying the stated Green's-function representation lemma to w at T , followed by its $L^1(\Omega)$ bound, gives

$$\|w(T)\|_{\infty, \Omega} \leq \Phi(0)\|w(0)\|_{\infty, \Omega} + \sum_{j=1}^M \int_{I_j} \Phi(s) \left(\|\delta_t \rho^j\|_{\infty, \Omega} + \|f(s) - f^j\|_{\infty, \Omega} + (t_j - s)\|\delta_t(f^j + \psi_h^j)\|_{\infty, \Omega} \right) ds.$$

Here $w(0) = u_0 - u_h^0 - \rho^0$. Since $\tilde{R}(T) = R^M$, the triangle inequality, the reconstruction estimates, and $u_h(T) = u_h^M$ therefore prove

$$\begin{aligned} \|u(T) - u_h(T)\|_{\infty, \Omega} &\leq \kappa_0 e^{-\gamma T} (\|u_0 - u_h^0\|_{\infty, \Omega} + \eta_{\text{ell}}^0) + \eta_{\text{ell}}^M \\ &\quad + \sum_{j=1}^M \eta_{\text{ell}, \delta_t}^j \int_{I_j} \Phi(s) ds \\ &\quad + \sum_{j=1}^M \int_{I_j} \Phi(s) \|f(s) - f^j\|_{\infty, \Omega} ds \\ &\quad + \sum_{j=1}^M \|\delta_t(f^j + \psi_h^j)\|_{\infty, \Omega} \int_{I_j} \Phi(s)(t_j - s) ds. \end{aligned}$$

The two terms on the last two lines are respectively the forcing-mismatch and interpolation indicators. This inequality is understood in the extended sense if an indicator is infinite. In general dimensions, $f \in L^2(0, T; H_0^1(\Omega))$ alone does not ensure that the displayed spatial maximum norms are finite; thus the stated assumptions do not, by themselves, yield a guaranteed finite maximum-norm estimator. Such a conclusion requires additional spatial boundedness sufficient to make those indicators finite.